

15. Keep rows with at least 2 NaN values

AIM

To display only the rows containing at least two missing values.

INPUT

A DataFrame with NaN values.

CODE

exp15.py > ...

```
1 import pandas as pd
2 import numpy as np
3 df = pd.DataFrame({
4     'A': [1, np.nan, np.nan, 4],
5     'B': [np.nan, 2, np.nan, 5],
6     'C': [3, np.nan, 6, 7],
7     'D': [4, 5, np.nan, np.nan]
8 })
9 print("Original DataFrame:")
10 print(df)
11 result = df[df.isnull().sum(axis=1) >= 2]
12 print("\nRows with at least 2 NaN values:")
13 print(result)
```

OUTPUT

Original DataFrame:

	A	B	C	D
0	1.0	NaN	3.0	4.0
1	NaN	2.0	NaN	5.0
2	NaN	NaN	6.0	NaN
3	4.0	5.0	7.0	NaN

Rows with at least 2 NaN values:

	A	B	C	D
1	NaN	2.0	NaN	5.0
2	NaN	NaN	6.0	NaN

RESULT

Thus, rows containing at least two NaN values were displayed successfully.

16. Split a DataFrame into groups based on school code and check the type of GroupBy object

AIM

To group a DataFrame based on school code and verify the type of the GroupBy object.

INPUT

Sample student DataFrame.

CODE

exp16.py > ...

```
1 import pandas as pd
2 data = {
3     'School_Code': ['S1', 'S2', 'S1', 'S3', 'S2', 'S1'],
4     'Student': ['John', 'Alice', 'David', 'Mary', 'Tom', 'Sara'],
5     'Marks': [85, 90, 78, 88, 76, 95]
6 }
7 df = pd.DataFrame(data)
8 print("Original DataFrame:")
9 print(df)
10 group = df.groupby('School_Code')
11 print("\nType of GroupBy Object:")
12 print(type(group))
13 print("\nGrouped Data:")
14 for name, grp in group:
15     print("\nSchool Code:", name)
16     print(grp)
```

OUTPUT

Original DataFrame:				Grouped Data:			
	School_Code	Student	Marks		School_Code	Student	Marks
0	S1	John	85		S1	John	85
1	S2	Alice	90	0	S1	David	78
2	S1	David	78	5	S1	Sara	95
3	S3	Mary	88		S2		
4	S2	Tom	76		S2	Alice	90
5	S1	Sara	95	4	S2	Tom	76
Type of GroupBy Object:					S3		
<class 'pandas.api.typing.DataFrameGroupBy'>				3	S3	Mary	88

RESULT

Thus, the DataFrame was successfully grouped by **School_Code**, and the GroupBy object type was verified as **pandas.core.groupby.generic.DataFrameGroupBy**.